

Three Worlds of 2041

How AI could reshape our industry, our customers and our planet over the next 15 years

THREE SCENARIOS, NOT THREE PREDICTIONS

On 6 April 2026, four people watched Earth set behind the Moon. Fifteen years from now, the Moon may be a place where people go to work. This is not a dream. It is three futures we can already see coming, and the choices that decide which one we get.

IN THIS BRIEFING today's signals, our method, three worlds of 2041, the odds, and what to do from Monday.



Mohammad

Senior Futures Consultant, Em, Oli & Timmonds

Part 1 | The Creative Solution

Earth sets behind the Moon, photographed by the Artemis II crew, 6 April 2026. Image: NASA

One story, fifteen steps

Our brief: imagine the world in 15 years' time, considering AI's impact on our industry and on the whole world, and decide what E,O&T should build now without betting on one forecast. The story uses an adapted, branching Story Spine: the three "Because of that" steps are alternative worlds. Facts and projections are tagged; the 2041 worlds are imagined scenarios.



SCOPE Industry: strategic consulting | Clients: energy, logistics and digital services | World: work, daily life, infrastructure and the space frontier

The future is already leaking in

Four signals we can measure in 2026, found with two of the 4 lenses of innovation: harnessing trends and challenging orthodoxies. Every world grows out of them.

2026 2041



415 TWh

used by data centres in 2024, about 1.5% of world electricity, with about 945 TWh by 2030 as a base-case projection. (IEA)

Energy becomes AI's ceiling.



170M new jobs

created and 92M displaced by 2030 across major economic and technological trends, including AI; 39% of core skills change. (WEF)

Skills, not jobs, become the scarce asset.



54,000

debris objects larger than 10 cm circle Earth, plus about 1.2 million between 1 and 10 cm. (ESA)

Orbit is a finite resource we can pollute.



April 2026

Artemis II flies four astronauts around the Moon, the first crewed lunar flight since 1972. (NASA)

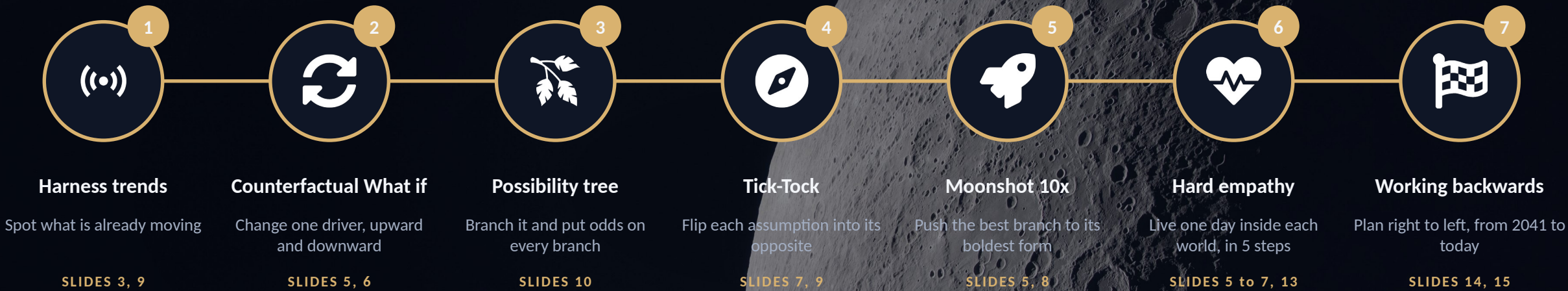
Space moves from exploration to economy.



Challenging orthodoxies. The old belief "AI is software" no longer holds. AI is an energy, materials and orbit industry, and that is exactly where our three worlds split apart.

The E,O&T Futures Engine

How the signals become worlds. Seven tools work like gears, and each produces a visible output on the slides shown. We first ran a controlled What if (keep AI access fixed, change only energy headroom), then combined both uncertainties into three worlds.



AI "bad copying"

We asked AI for dozens of variants of each world and kept the strange ones. Its mistakes became raw material.

Exaptation

Borrowed from air traffic control (one network, many vehicles) and multiplayer game design (rules that shape behaviour).

Shared, verified AI with abundant clean energy



Robots on the Moon (NASA concept)



A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Layla, 34, grid engineer, Riyadh

Scene: 6 am. My AI agent has already sold my solar surplus and drafted my report.

Feel: free, and strangely idle.

Make sense: the machines now do the routine half of my job.

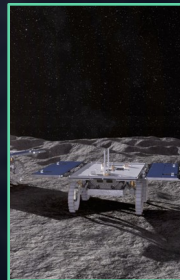
Worry: what am I still for?

Act: I join the council that designs our city's digital twin.

A future we want to be part of

Mirror Cities

Customers try a flat or a lunar hotel in a live digital twin before buying. White-hat Octalysis: each purchase plants a real tree.



Space and the Moon

A crewed base at the lunar south pole mines ice for fuel. Robots build; the first tourists arrive.



Life and transport

Electric pods and air taxis on one AI network, borrowed from air traffic control. Car parks become parks.



Our industry

AI writes the analysis; consultants own the judgement. E,O&T move: launch a future-design practice.

5 Cs

Character

Layla, and every
client like her

Conflict

AI's hunger for energy

Cure

Clean power, efficient chips

Change

Abundance, shared widely

Carry-out

Build the future we
want

Still hard here: e-waste, attention overload, disputes over lunar mining.

Concentrated, opaque AI with scarce, polluting energy



Hardware in low orbit (NASA, ISS)



Omar, 41, logistics manager

Scene: in the depot, two delivery alerts contradict each other.

Feel: anxious; the rationing app also limits when I can charge the vans.

Make sense: one of the alerts may be synthetic.

Worry: I could send a driver onto an unsafe route.

Act: I verify both alerts through our data cooperative before dispatch.

STORY SPINE Because of that (2)

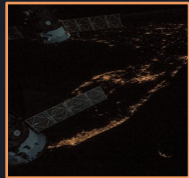
The honest shadow of World 1. Same engine, downward What if, told as a chain of cause and effect.

1

AI demand outran clean power, so fossil plants kept running.



The metaverse is an escape; black-hat Octalysis (scarcity, loss) keeps people hooked.



Space and the Moon

Low orbit is a junkyard; the lunar base waits.



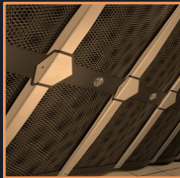
Heat and pollution rose,
and power was rationed.



A few players owned the models; deepfakes broke trust.



Empty robotaxis gridlock the cities.
The rich fly; everyone else waits.



Our industry

Reports cost pennies. E,O&T move:
verification and resilience services.



A debris cascade took out navigation and broadband satellites.

5 Cs

Character
Omar

Conflict

AI outgrows clean power

Cure

None, in time

Change

Trust and orbit lost

Carry-out

Plan for the shadow first

PATCHWORK

2041

Tick-Tock: the middle path

Trust verified inside regional blocs, mixed energy between them

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Trust verified inside regional blocs, mixed energy between them

One planet, three rulebooks.



 A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Sara, 29, product designer, Stockholm

Scene: my AI agent is doing my weekly shopping.

Feel: relaxed, until it refuses my favourite app.

Make sense: that app has no trust seal in our bloc.

Worry: my own product could be locked out the same way.

Act: I design one product that passes all three blocs' trust rules.

Neither heaven nor collapse

The middle path, built by flipping two of today's facts with Tick-Tock. Here trust, not technology, decides who wins.

TICK People buy from companies.

TOCK AI agents buy from AI agents, so firms must design for machine customers.

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TICK Data is free.

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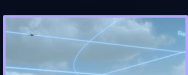
A person wearing a VR headset and a white shirt with a black harness, looking at a virtual environment displayed on a screen. The screen shows a blue and white structure, possibly a ship or a large building, with a red and white striped pole in the foreground. The background is dark and indistinct.

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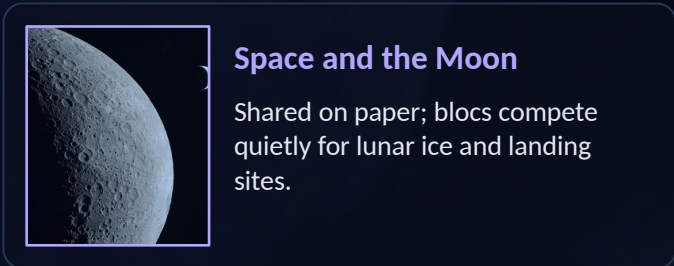
Life and transport

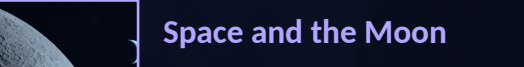
Driverless in some regions, manual in others; the best cities blend both.



Life and transport

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Space and the Moon

Shared on paper; blocs compete quietly for lunar ice and landing sites.

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Our industry

Winners translate between blocs.
E,O&T move: a cross-bloc standards advisory.



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5 Cs	Character Sara	Conflict One world, many rulebooks	Cure Shared trust seals	Change Identity is the product	Carry-out The future is a negotiation
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From your door to orbit

Borrow the delivery-network logic, not the aircraft: a model that brings dinner today, adapted for orbital logistics, could help feed astronauts in 2041.

2026

OBSERVED



Drones deliver food now

One delivery network passed 2 million commercial drone deliveries in January 2026, with a median flight of 3 minutes (Food On Demand).

TRANSFER: scheduling, tracking, exception handling

2030s

PROPOSED TEST



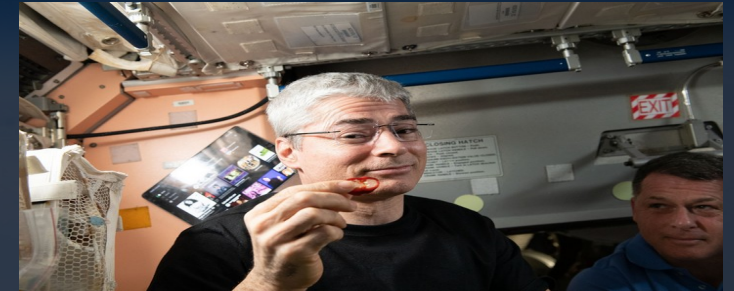
Autonomy reaches orbit

Robotic freighters already berth at the station using its robotic arm. Next: test AI scheduling with qualified space partners.

GATE: safety, communication delay, payload limits

2041

MOONSHOT



Feeding the crew in space

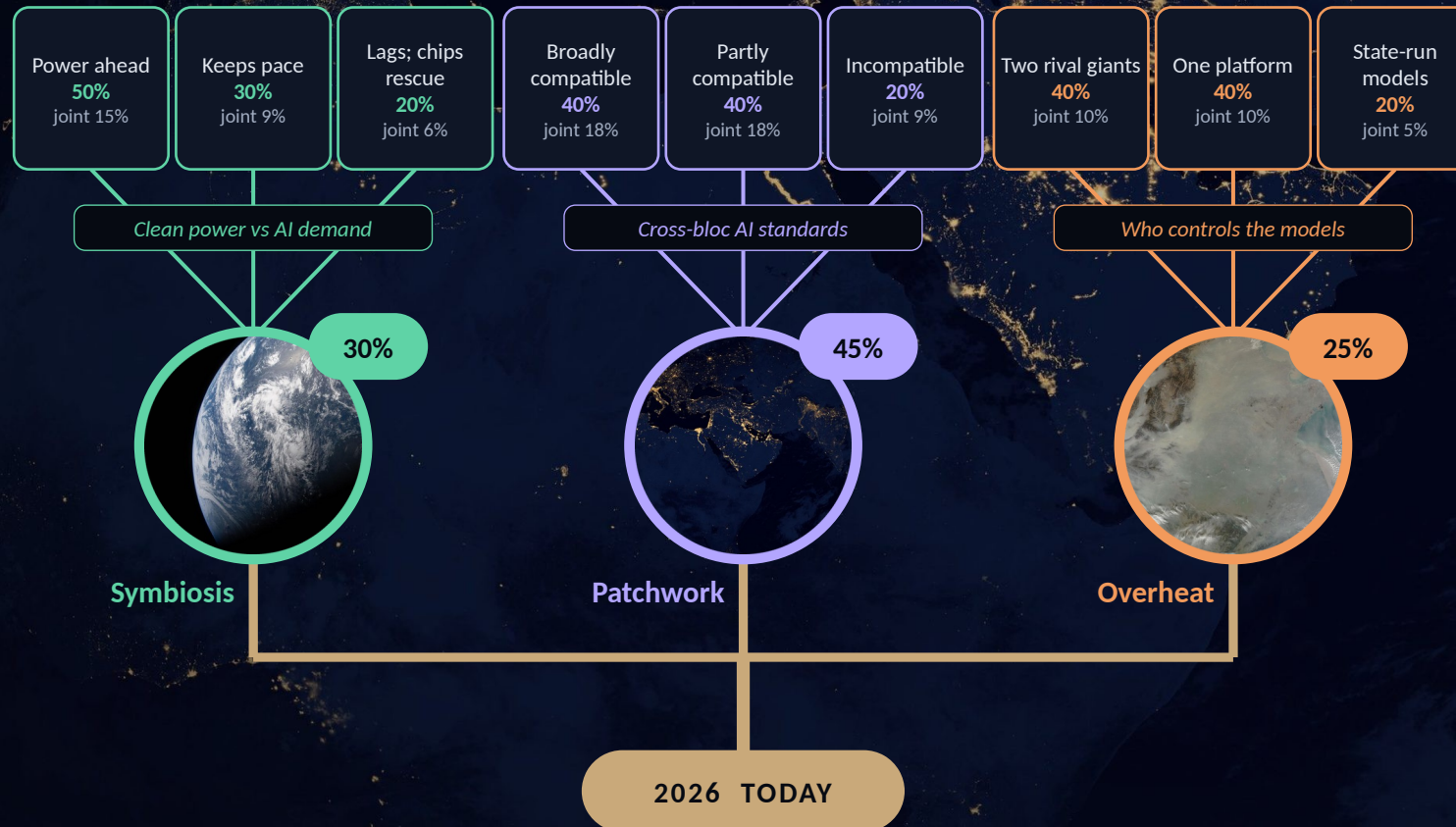
Each astronaut needs about 1.7 kg of food a day; fresh produce arrives every three months (NASA). Imagine cargo and orbital crops supporting a lunar outpost.

10x AMBITION: on-demand, not occasional, resupply

E,O&T test. Start with a logistics simulation and a partner review, not launch infrastructure. For clients in logistics and food, the last mile becomes the last orbit.

Which branch are we on?

Three worlds with planning weights, then one key driver per world split into three exclusive outcomes. Conditional odds sum to 100% per world, and the joint odds of all nine outcomes sum to 100%. Illustrative judgements, not calibrated forecasts.



How to read the tree

Level 1: worlds

Planning weights for the three worlds: 30%, 45%, 25%. They sum to 100%.

Level 2: one driver each

Three mutually exclusive outcomes per world, with conditional odds that sum to 100% within that world.

Joint odds

World x outcome. Worked example: Patchwork and partly compatible = $45\% \times 40\% = 18\%$. All nine joint odds total 100%.

Why Patchwork leads

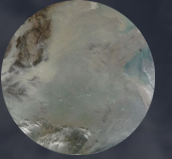
AI rules are already splitting by region; the clean-power race is not yet won.

Status

Illustrative judgements to support discussion, not calibrated forecasts.

One causal story inside one world

The worlds branch, but a story inside a world must follow cause and effect. All eight Story Spine steps for Omar in Overheat, a fictional rehearsal.



01 ONCE UPON A TIME

Omar ran a depot whose customers depended on daily deliveries.



02 EVERY DAY

His team dispatched the vans on the AI's recommendation.



03 BUT ONE DAY

Two alerts conflicted, and the planned van could not charge.



04 BECAUSE OF THAT

Omar paused the recommendation before sending a driver.



05 BECAUSE OF THAT

He checked the records and called the depot to verify the route.



06 BECAUSE OF THAT

He used the manual fallback and
logged why the decision changed.



07 UNTIL FINALLY

A verified alternative kept the delivery moving.



08 AND EVER SINCE THEN

The team reviews every exception with a Decision Passport.

Takeaway (5 Cs carry-out). The cure is not perfect AI. It is a process that keeps evidence, ownership and a fallback visible.

Prove it is useful before scaling it

One client, one workflow, one limited rehearsal. All activities and gates below are proposed; no client commitment or benefit is claimed.

DAYS 1 TO 30



Define and observe

OWNERS Client owner and consultant

Map one workflow, record a manual baseline and interview five users.

GATE: agree success, error severity, data permissions and limits.

DAYS 31 TO 60



Prototype and rehearse

OWNERS Technical lead and client users

Build a simulation and a Decision Passport; rehearse all three worlds.

GATE: five users can name the evidence, owner and fallback.

DAYS 61 TO 90



Stress-test and decide

OWNERS Sponsor and independent reviewer

Run ten scripted cases: low power, false input, provider outage and normal operation.

GATE: every critical fallback works; no unresolved material error.

Continue, change or stop. Scale only when the pilot shows client-valued benefit and acceptable resource use.

Pilot limit. Five users and ten cases support learning, not proof of general reliability.

**The Moon is closer than ever.
So is 2041.**

Three worlds, one discipline: make the assumptions visible, test the response, and keep a human decision owner in the loop.

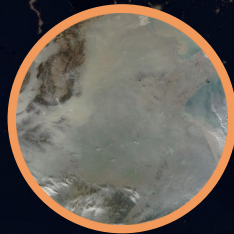


Symbiosis
Build it



Patchwork

Shape it



Overheat
Insure against it

We do not predict the future. We prepare for all three, and we work every day for the first one.

Mohammad, Senior Futures Consultant, Em, Oli & Timmonds



What I am taking with me

The tools that resonated, the change in my thinking, and why it matters in my professional life.

Three ideas from this course changed how I think about planning: possibility trees, working backwards, and counterfactual thinking. Together they gave me a structured language for something my work has always demanded: decisions about a future nobody can see.

In twenty years of network engineering and digital transformation, I have learned that a platform chosen today may still shape a business in ten years. Building the three worlds of 2041 for this assignment showed me a different way to approach that kind of decision. Drawing the possibility tree forced me to give each branch a probability, and the less likely branches raised questions a single forecast would never have asked. Working backwards from a 2041 press release, the way Amazon writes the press release before the product exists, made the first steps for 2026 obvious. Counterfactual thinking kept the exercise honest: writing Overheat next to Symbiosis stopped me from falling in love with the optimistic future.

What resonated most is that creativity here is not a talent but a structured skill, wild ideas plus strategy.



20 years

in networks and digital transformation,
where decisions have long consequences

How I will put it into practice

A realistic timetable, a review loop and a personal routine that continues beyond this course.

My plan is concrete. From November, every new technical proposal I write will open with a one-page working backwards brief describing the result for the customer in three years, the assumptions behind it and the first test. Once a quarter, starting in January 2027, I will build a possibility tree for one roadmap decision, with three branches, nine sub-branches and probabilities, and review it three months later to see which branch reality followed. Before each major design review I will run one upward and one downward counterfactual with the team and record the risks it surfaces.

Personally, I live between Riyadh and Stockholm, and I will turn those flights into fuzzy focus time: no screens, just my stimuli queue and a commonplace book, so slow hunches have somewhere to land.

I will measure success simply: by June 2027, I will have built two possibility trees, completed the first three-month review, and recorded whether the findings changed a roadmap decision.



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Image credits

Slide 1: "A Breathtaking Earthset from Orion", Artemis II crew, 6 April 2026. NASA (art002e021278).

Slide 3: Night Earth observation from the ISS, Expedition 40. NASA (iss040e085126).

Slides 2, 4 and 14: "Home in the Distance", Artemis II. NASA (art002e020922).

Slide 5: "Earth in View", Artemis II (art002e023506); CADRE rovers concept, NASA/JPL-Caltech (PIA26296).

Slide 6: "Haze over eastern China", NASA Goddard (GSFC_20171208_Archive_e001269); ISS Expedition 35 (iss035e017673).

Slides 7, 10 and 17: "Black Marble: Africa, Europe, and the Middle East", NASA Goddard (GSFC_20171208_Archive_e001591).

Slides 5 to 7 and 16: NASA VR training (jsc2026e000427), Advanced Air Mobility concepts (AAM-Concept-2021, automation concept), Mission Control (jsc2026e019925), Pleiades supercomputer (ARC-2008-ACD08-0272-005).

Slide 8: NASA drone flight test (AFRC2026-0152-22); Cygnus in Canadarm2 (iss071e418235); Expedition 66 chile peppers (iss066e023179).

Slide 9: ISS water recovery (iss072e143163); sand dunes and irrigation in Saudi Arabia (PIA11069); Earth limb and Moon (iss051e039977); Veg-01 harvest (iss040e009126); JWST golden mirror (GSFC_20171208_Archive_e000368).

World thumbnails on slides 2, 4, 10, 12, 13 and 17 reuse the images of slides 5, 6 and 7.

All NASA images via the NASA Image and Video Library, <https://images.nasa.gov> (public domain).

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Use of AI

I acknowledge the use of Claude by Anthropic (<https://claude.ai>) and ChatGPT by OpenAI (<https://chatgpt.com>) to help generate and develop scenario ideas and variants, critique and develop wording, framework applications, scenario logic and probabilities, design the Decision Passport, pilot and workshop, find and check sources, and arrange and format the slides. The prompts used include "Propose solution directions for each company and the course concepts they would apply", "Draft three different future scenarios for 2041 with personas, built on the course tools", "Check every slide against the assignment requirements and rubric", "Merge the strongest ideas from my two drafts into one deck" and "Build the deck as a PowerPoint with my photo and a dark cinematic layout". The output was used as a starting point that I selected, reviewed, edited and verified; the choice of company and direction, the final scenarios, and the reflection based on my own experience and plans are mine, and I remain responsible for the submission.